

ICT E-LEARNING PLATFORM

-Developed by:- Jash Jani

This project report provides an overview of the E-Learning Management System (LMS) developed for the Marwadi University FODS-ICT department. The platform is designed to facilitate seamless interaction between faculty and students for course delivery and progress tracking.

1. Project Overview

The E-Learning Platform is a web-based application that allows faculty members to upload educational content (videos and study materials) while providing students with a centralized portal to access resources, track their learning progress, and earn certificates upon completion.

2. System Architecture

The application follows a Monolithic Architecture using the Flask web framework. It handles routing, user authentication, and file management within a single unified service.

Technical Stack

Backend: Python (Flask)

Database: * Volatile Memory: In-memory dictionary for session-based data handling.

Persistent Storage: SQLite (as defined in init_db.py) for user and progress records.

Frontend: HTML5, CSS3 (embedded styles), and Vanilla JavaScript for asynchronous updates (AJAX).

PDF Generation: ReportLab library for dynamic certificate creation.

Storage: Local file system for uploaded media and documents.

3. Key Features

A. User Roles & Authentication

The system supports two distinct roles with specific permissions:

1. Student: Can view videos, download materials, and track completion percentage.
2. Faculty: Can upload, rename, and delete course content.

B. Content Management

Faculty can manage two types of educational assets:

1. Videos: Integrated with a custom "Watch" tracker.
2. Materials: General file support (PDFs, PPTs, etc.) for reading resources.

C. Progress Tracking & Certification

Asynchronous Tracking: Using JavaScript's fetch API, the system marks a video as "Watched" the moment a student hits the play button without refreshing the page.

Dynamic Progress Bar: A visual indicator calculates completion based on:

$$\text{Progress\%} = (\text{Total Videos} - \text{Videos Watched}) \times 100$$

Automated Certification: Once progress reaches 100%, a modal is unlocked allowing students to generate a customized PDF certificate.

4. Database Schema

The system utilizes a relational database structure (lms.db) to maintain data integrity:

Modules	Purpose
Users	Stores credentials (Enrollment/Username, Password/GR) and roles.
Videos	Maintains a registry of uploaded video titles and their server paths.
Progress	A junction table mapping student IDs to specific video IDs to track completion.

5. File Structure

- app.py:** The core application logic, route handlers, and UI templates.
- init_db.py:** Database initialization script for setting up tables and seed data.
- /uploads:** A secure directory for storing physical media files.
- /templates:** These files serve as dynamic templates that use Jinja2 syntax to render user-specific interfaces, such as the faculty upload form or the student video gallery, based on the logged-in user's role.
- script.js:** This file manages the asynchronous logic of the platform by fetching video data, updating the completion percentage text, and notifying the server to save progress whenever a video finishes playing.

6. Security Implementation

Session Management: Uses a secret_key to sign cookies, preventing unauthorized session spoofing.

- File Security:** Utilizes werkzeug.utils.secure_filename to prevent directory traversal attacks during file uploads.

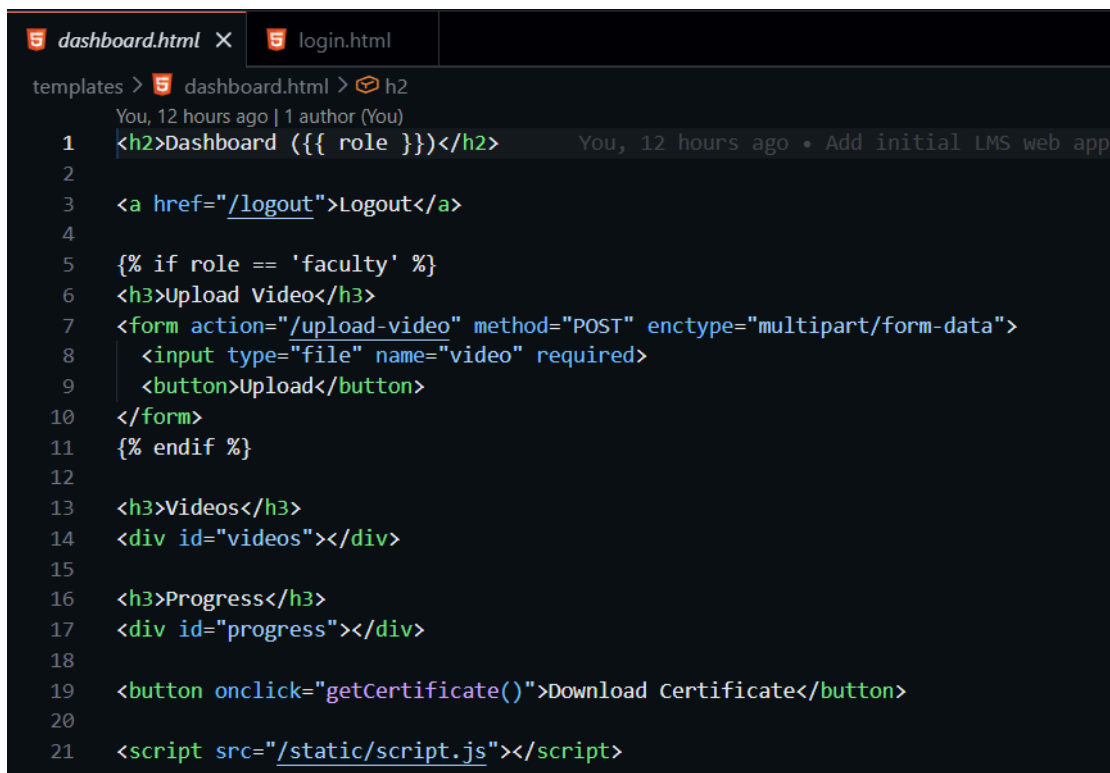
- b. **Access Control:** Route-level decorators check session.get('role') before allowing faculty actions like delete_content.

7. Conclusion

This platform provides a robust foundation for departmental e-learning. Its lightweight design ensures high performance, while the integration of automated PDF generation adds professional value for the student users. Future enhancements could include persistent storage for the mock data dictionary and a more advanced video playback tracker based on time-duration.

SCREENSHOTS

1. Codebase



```
dashboard.html x login.html
templates > dashboard.html > h2
You, 12 hours ago | 1 author (You)
1 <h2>Dashboard ({{ role }})</h2>
2
3 <a href="/logout">Logout</a>
4
5 {% if role == 'faculty' %}
6 <h3>Upload Video</h3>
7 <form action="/upload-video" method="POST" enctype="multipart/form-data">
8   <input type="file" name="video" required>
9   <button>Upload</button>
10 </form>
11 {% endif %}
12
13 <h3>Videos</h3>
14 <div id="videos"></div>
15
16 <h3>Progress</h3>
17 <div id="progress"></div>
18
19 <button onclick="getCertificate()">Download Certificate</button>
20
21 <script src="/static/script.js"></script>
```

```
<form method="POST">
  <h2>Login</h2>

  <input name="username" placeholder="username" required>
  <input name="password" type="password" placeholder="password" required>

  <button type="submit">Login</button>
</form>
```

```
// Load videos
fetch("/videos")
.then(res => res.json())
.then(data => {
  const container = document.getElementById("videos");

  data.forEach(v => {
    const vid = document.createElement("video");
    vid.src = v.path;
    vid.controls = true;
    vid.width = 300;

    vid.onended = () => {
      fetch("/watch/" + v.id, { method: "POST" })
        .then(() => loadProgress());
    };

    container.appendChild(vid);
  });
});

// Load progress
function loadProgress() {
  fetch("/progress")
    .then(res => res.json())
    .then(data => {
      document.getElementById("progress").innerText =
        "Completion: " + data.progress.toFixed(2) + "%";
    });
}

loadProgress();

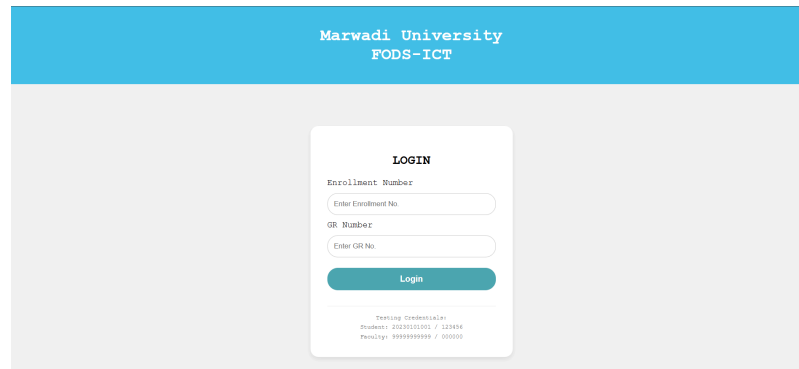
// Certificate
function getCertificate() {
  window.location.href = "/certificate";
}
```

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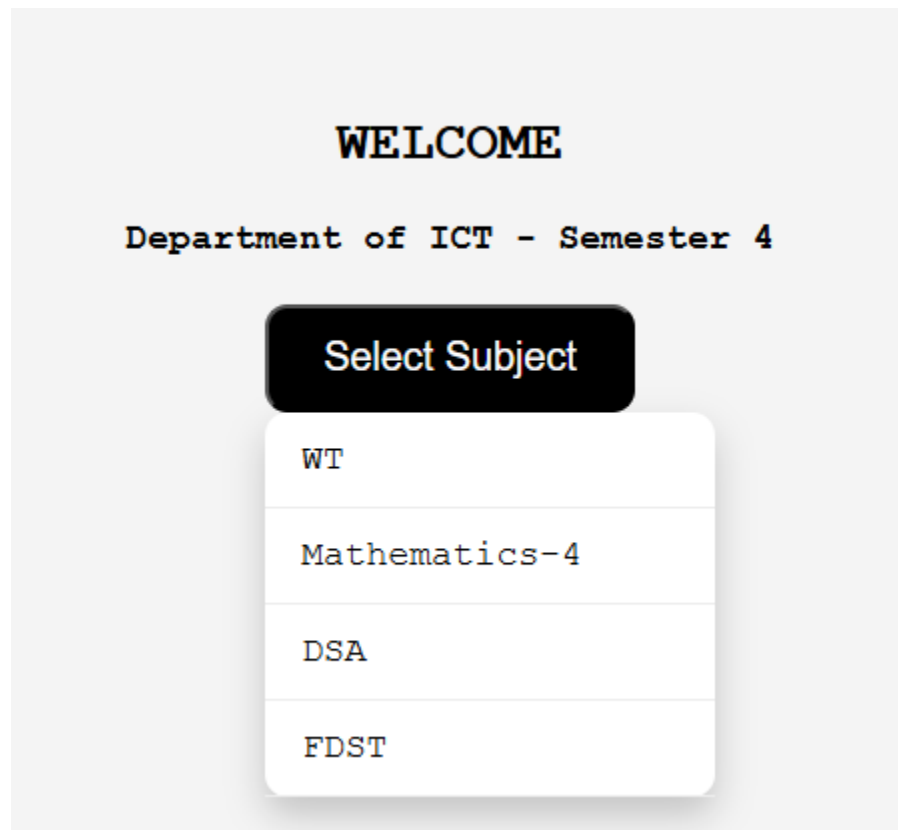
2. PROJECT-INTERFACE

a. Login interface



The login interface features a blue header with the text "Marwadi University" and "FODS-ICT". Below the header is a white login form with a teal "Login" button. The form includes fields for "Enrollment Number" and "GR Number", each with a placeholder "Enter Enrollment No." and "Enter GR No." respectively. At the bottom of the form, there is a small text block: "Testing Credentials: Username: 2020010101 / 12345 Password: 99999999 / 00000".

b. Subject Catalogue



The subject catalogue interface displays a "WELCOME" message followed by "Department of ICT - Semester 4". Below this, there is a black button labeled "Select Subject". A white dropdown menu is open, showing a list of subjects: "WT", "Mathematics-4", "DSA", and "FDST".

b. Faculty dashboard

[← Back to Subjects](#)

Mathematics-4

Videos

Materials

Course Videos

Video Title

Choose File

No file chosen

Upload Video

Study Materials

Material Title

Choose File

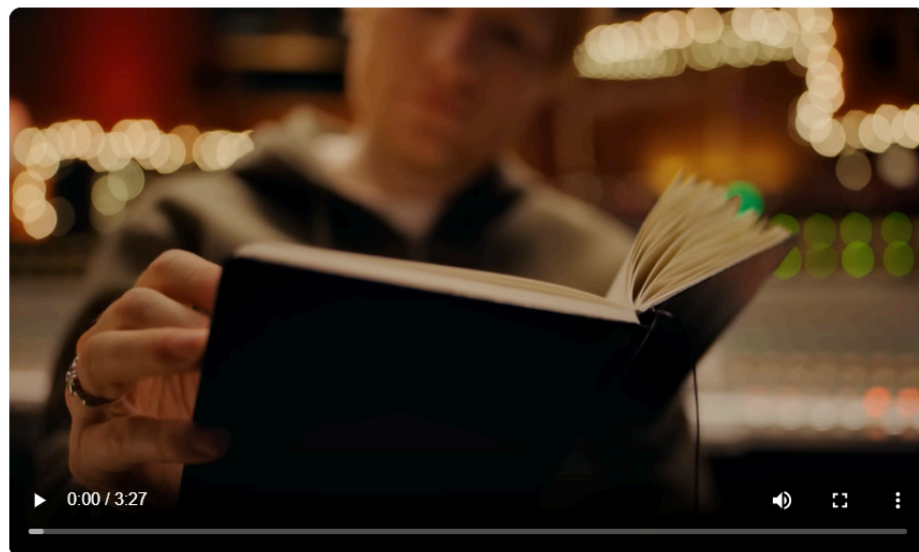
No file chosen

Upload Material

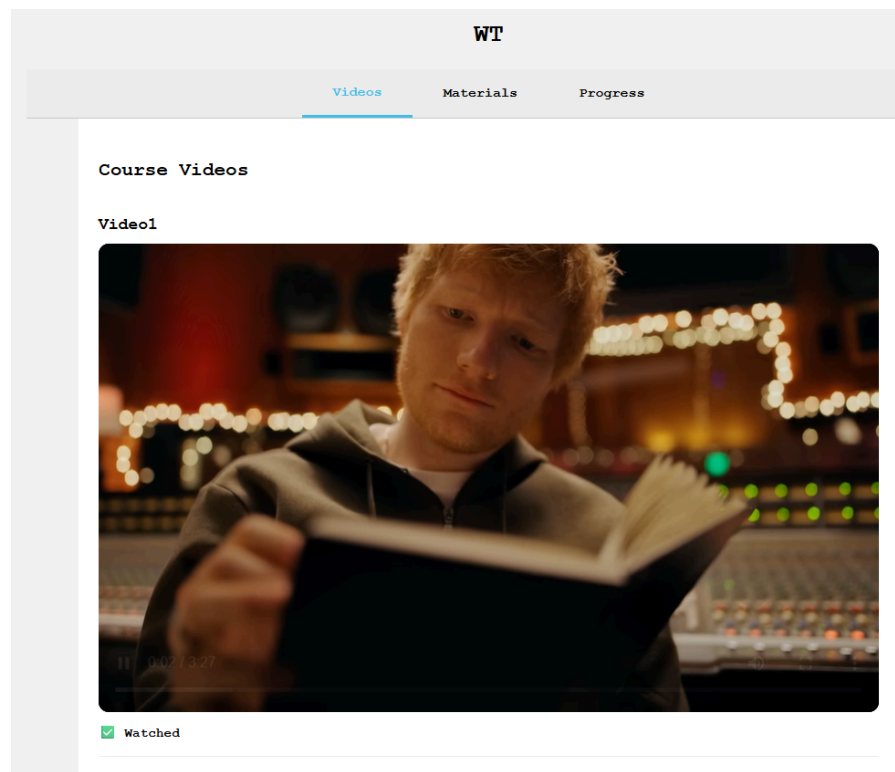
Video1

Edit

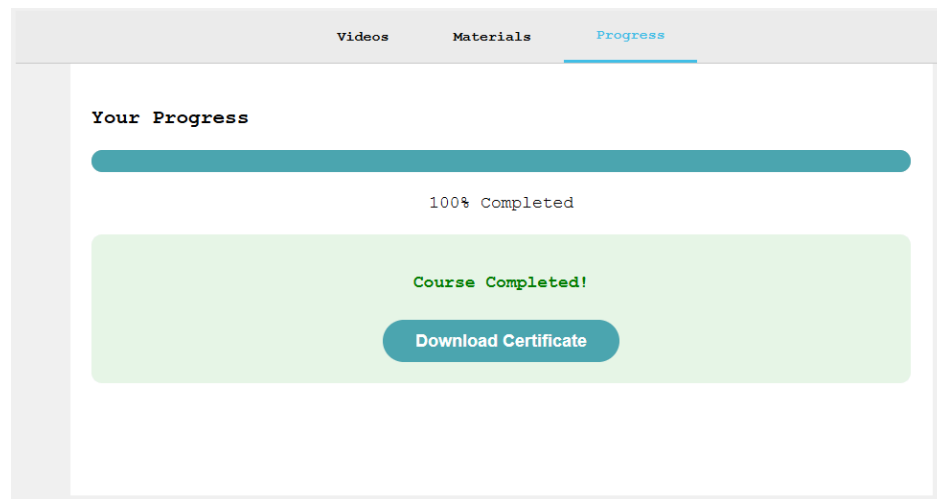
Delete



c. Student Dashboard



Course completion



Certificate details

Certificate Details

Please confirm your details for the certificate.

Full Name

Jash Jani

Confirm Enrollment

92400957004

Cancel

Generate PDF

Course completion certificate pdf file

CERTIFICATE OF COMPLETION

This is to certify that

JASH JANI

Enrollment: 92400957004

has successfully completed the course

WT

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